

Applied A.I. Solutions

Foundations of Data Management (T431)

Data Governance

*Exercises 1, 2, 3
and
Final Project*

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Objective under the Data Governance Framework

The goal is to establish a structured process for extracting, transforming, and loading (ETL) GBP_DataSource Business Data into a governed environment, ensuring accuracy, consistency, and compliance with data standards. The processed data will serve as the foundation for generating two recurring outputs:

1. **Operational Report** – focused on day-to-day business metrics, enabling managers to monitor sales, inventory, and customer transactions in near real-time.
2. **Executive Report** – providing high-level strategic insights, such as profitability trends, regional performance, and customer segments, tailored for organizational leadership.

This approach ensures that business data is not only integrated and reliable but also aligned with the organization's **data governance principles** of data quality, accountability, and accessibility.

Learning Outcomes

By completing this assessment, students will be able to:

1. Business Analysis & Reporting

- Analyze business datasets to identify key metrics, trends, and stakeholder needs.
- Differentiate between operational and executive reporting requirements.
- Design report structures that support decision-making at multiple organizational levels.

2. Data Quality & Governance

- Assess and improve data quality through cleaning, validation, and transformation.
- Apply data governance principles to ensure accuracy, consistency, and compliance.
- Document metadata and lineage to support transparency and accountability.

3. Data Modeling & Architecture

- Develop Data Flow Diagrams (DFDs) to map business processes and data movement.
- Create Entity-Relationship Diagrams (ERDs) using normalization techniques (2NF/3NF).
- Distinguish between master data, reference data, and transactional data.

4. ETL Implementation

- Execute ETL processes using tools such as Excel, SQL, and Python.
- Build and populate reporting tables in MySQL based on a logical schema.
- Automate and validate ETL workflows to ensure reliable data delivery.

5. Governance Framework Design

- a. Apply the DAMA Wheel sub-domains to a real-world reporting pipeline.
- b. Define roles and responsibilities for data stewardship and ownership.
- c. Identify and mitigate risks related to data mismanagement and reporting errors.

6. Communication & Documentation

- Produce professional documentation including business reports, technical logs, and governance frameworks.
 - Present findings and insights in a format suitable for both technical and executive audiences.
 - Collaborate effectively within teams to deliver structured, high-quality outputs.
-

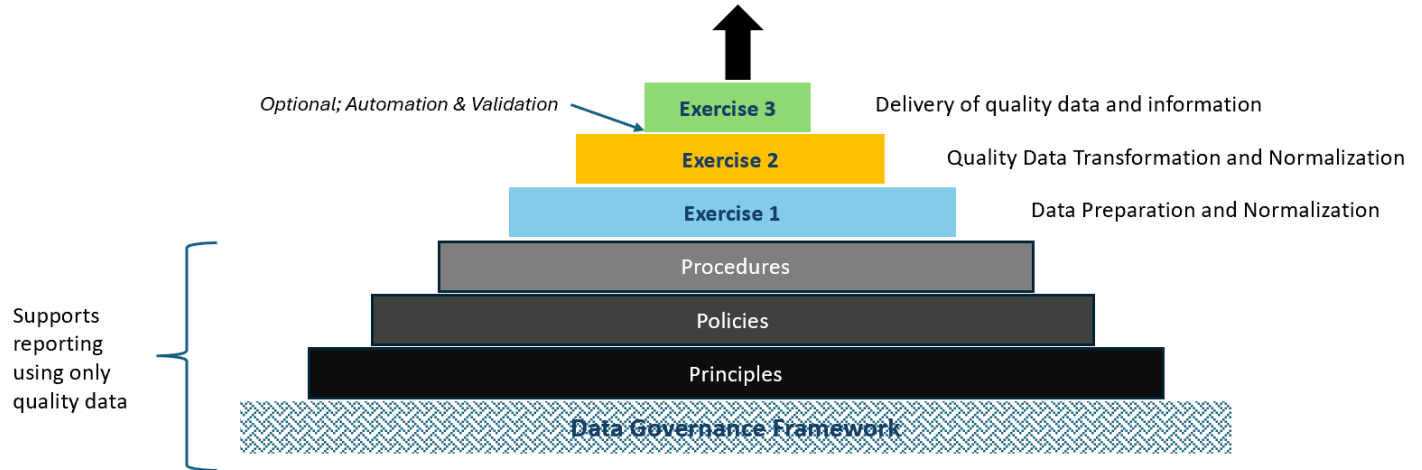
Rubric

- Completeness of Deliverables (3 pts)
- Accuracy of Analysis and Modeling (3 pts)
- Clarity and Documentation (2 pts)
- Governance Alignment and Insight (2 pts)

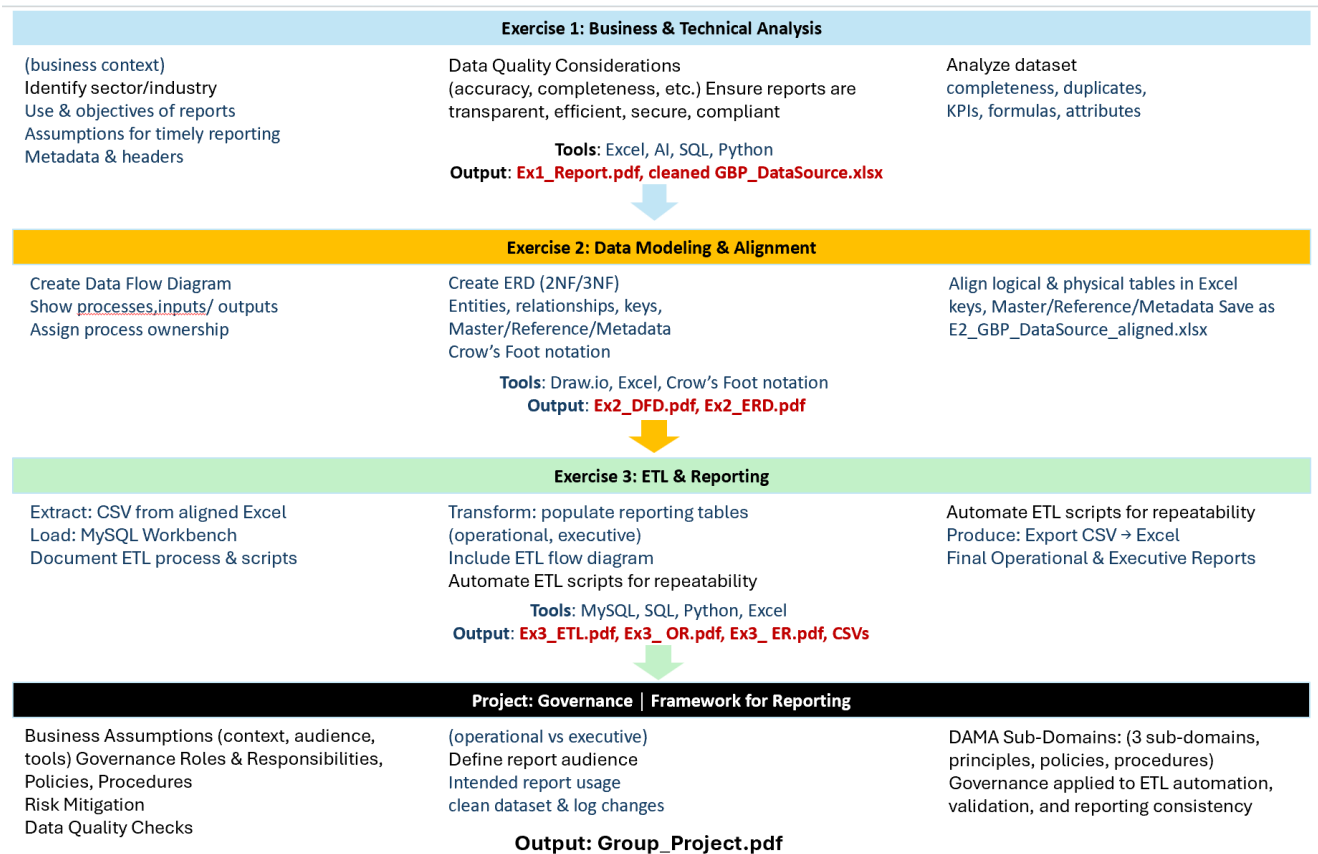
Data Governance Exercises + Project Blueprint

Exercises + Project Data Governance Blueprint

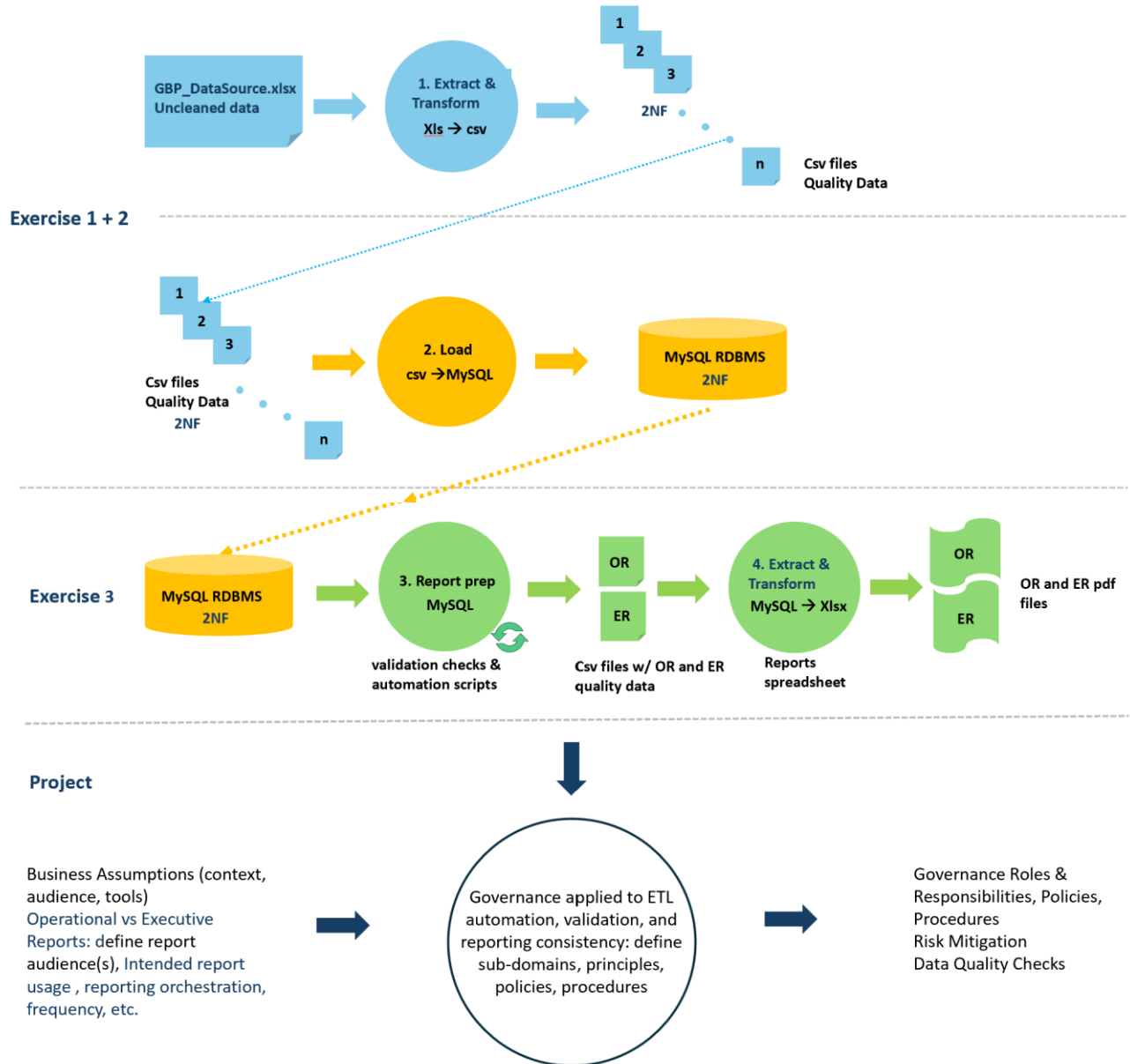
- Decision Making
- Performance Improvement
- Analysis, Research and Development
- Process Monitoring and Control
- Communication
- Training



Project Charter



Data Flow Diagram



Lab Exercise 1: Business and Technical Analysis – 10 points max

Part 1 – Business Context

Analyze the *GBP_DataSource* dataset. Prepare a written report addressing the following:

- Identify the sector/industry of study.
 - Define the target audience for each report (operational vs. executive).
 - Specify the intended use of each report (e.g., monitoring & control, decision-making, performance improvement, research & analysis).
 - Describe the publishing and distribution frequency, as well as report maintenance and expected usage.
 - State the assumptions required to produce both reports in a timely and reliable manner.
 - List the information content (metadata) included in both reports (e.g., sales, geography, customers, revenue, KPIs). (*Refer to Annex II and Annex III, pp. 5–6.*)
 - Propose suitable headers (titles) for both reports.
-

Part 2 – Technical Data Analysis (*GBP_DataSource*)

Perform a structured analysis of the dataset and document your findings:

- Assess data quality:** completeness, inconsistencies, redundancies, and duplicates.
- Identify and define any **additional calculations** (formulas, algorithms), KPIs (ratios, metrics), new entities, or attributes you plan to include in the report dataset.
- Clean the dataset** for further use and analysis. Keep a detailed log of all modifications (updates, deletions, overrides, transformations).

Note: Groups may use AI, SQL, and/or Python code to perform the technical analysis. Submit supporting code or scripts if applicable.

(*) Important: students who do not have SQL experience can read SQL basics and self-study using the content on D2L. Students who know SQL may share with and transfer their knowledge to classmates.

Lab Exercise 2: Business Data Analysis and Data Modeling – 10 points max

Considerations

- it builds on Exercise 1's business and technical analysis and moves into modeling and alignment, which is how data governance, reporting, and data management are articulated, sequenced and done in practice.
 - It covers the full data lifecycle — from business process mapping (DFD) to data modeling (ERD) to implementation alignment (cleaned dataset).
 - Enforces best practices: 2/3NF, master/reference data distinction, metadata documentation.
 - Hands-on tools: Draw.io for diagrams, Excel for aligned data, and optional SQL/Python.
-

Part 1 – Business Data Analysis & Modeling

Using the cleaned *GBP_DataSource.xls* dataset from Exercise 1:

a. Data Flow Diagram (DFD)

- Identify business processes involved in reporting.
- Show data flowing into and out of each process.
- Indicate relationships among processes.
- Assign process ownership (who is responsible).
- Illustrate orchestration (how processes interact).

Tool: Draw.io

b. Entity-Relationship Diagram (ERD)

- Define entities, attributes, domains, relationships.
- Assign primary keys and foreign keys.
- Create Master Data tables (e.g., Customers, Products) and Reference Data tables (e.g., Regions, Categories).
- Include a Metadata table (definitions, data types, sources).
- Use Crow's Foot notation and normalize to 2nd/3rd Normal Form.

Tool: Draw.io

c. Logical-to-Physical Alignment

- Create new Excel tabs to align physical tables with the logical ERD.
 - Split the cleaned data source accordingly (e.g., Customers, Orders, Products).
 - Save the aligned dataset as *E2_GBP_DataSource_aligned.xlsx*.
-

Part 2 – Submission Requirements Exercise 1 + Exercise 2

a. Business + Technical Report

- Document business/technical findings from Exercises 1 + 2.
- Include a Cover Page (group #, team members).
- Attach SQL and/or Python scripts used for cleaning.
- Save as Ex1+Ex2_Report.pdf.

b. DFD

- Export as PDF and submit as Group#_Ex1_Ex2_DFD.pdf.

c. ERD

- Export as PDF and submit as Group#_Ex1_Ex2_ERD.pdf.

d. Aligned Data Source

- Submit as E2_GBP_DataSource_aligned.xlsx.
-

Exercise 3: ETL and Reporting – 10 points max

Considerations

- It closes the loop from conceptual (Exercise 1) → logical/physical (Exercise 2) → implementation with ETL and reporting (Exercise 3).
 - It mirrors how data governance and Business Intelligence pipelines are done in practice. Hands-on progression: go through ETL to implement your ERD in MySQL.
 - Tool integration: MySQL Workbench, CSV imports, SQL/Python scripting, Excel formatting.
 - Governance alignment: data lineage (ETL flow diagram), metadata, cleaning, and scripts.
 - Data Quality: automation (optional) ensures timeliness, consistency, and reliability.
 - Note: groups may use AI, SQL, and/or Python
-

Part 1 – Setup

- Install MySQL Workbench on your computer. *(See Annex for guidance.)*
-

Part 2 – Database Schema

- Using your ERD (2NF/3NF) from Exercise 2, create the physical schema in MySQL named GBP_DataSource.
-

Part 3 – ETL and Report Generation

Follow the steps below:

a. Extract

1. From E2_GBP_DataSource_aligned.xlsx, export each tab (table) as a separate CSV file.
2. Ensure data is complete, cleaned, and free from hidden characters.

b. Load

3. Use MySQL Workbench Table Data Import Wizard to load each CSV file into the corresponding MySQL table.

c. Transform

4. Create two reporting tables in MySQL:
 - operational_report
 - executive_report

Note: use SQL to include all additional attributes needed to complete each report.

5. Write SQL queries to populate these reporting tables from the base tables.
6. Optionally, add validation checks (e.g., row counts before vs. after load, null checks, data type conformity, foreign key enforcement).

d. Produce

7. From MySQL, export both reporting tables as CSV files:
 - operational_report.csv
 - executive_report.csv
 8. Open each in Excel and apply light formatting to produce the final Operational Report and Executive Report.
-

Part 4 – Documentation

- Create an ETL Report documenting the process step by step (use numbered bullets).
 - Add a customized ETL flow diagram (showing lineage: data source → staging → reporting tables → final reports).
 - Include all SQL/Python scripts used.
 - Optionally, groups may write Python or SQL scripts to automate ETL steps (e.g., automated CSV loading, scheduled transformations, report table refresh). Submit these scripts along with the ETL report.
-

Part 5 – Submission

1. ETL Report: Group#_Ex3_ETL.pdf (with cover page and contributors).
 2. Operational Report: Group#_Ex3_OR.pdf.
 3. Executive Report: Group#_Ex3_ER.pdf.
 4. CSV files: operational_report.csv, executive_report.csv.
-

Project: Governance Framework for Reporting

(Max: 10 points x 3; “**4–6 pages max**” —diagrams can be included in addition to text if needed)

Students must follow the guidelines outlined in the PDF document posted on D2L/Assignments:

“DG Principles Policies Procedures - Guidelines (Self-study).pdf”

Considerations

- ✓ This Project is the culmination of the three lab exercises: from hands-on data modeling (Labs 1–3) to theoretical governance application (DAMA Wheel).
 - ✓ It helps to understand how data management frameworks guide practice.
 - ✓ It shows a clear alignment with the DAMA Wheel (theory → practice).
 - ✓ It shows structured deliverables: business assumptions, data quality considerations, sub-domains, principles → policies → procedures.
 - ✓ Students must apply governance theory to their own reporting process — not just restate definitions.
 - ✓ It uses the reporting process as the use case to ensure students tie people, process, and risk into the technical work already done.
-

A. Business Assumptions

1. Briefly describe the business context (GBP_DataSource dataset).
 2. State key assumptions about:
 - Business and stakeholders
 - Target audience of reports (operational vs. executive)
 - Technology used (ETL tools, SQL, Python, Excel, etc.)
 - Objectives of the reports (e.g.: monitoring, decision-making, performance improvement, etc.)
-

B. Data Quality Considerations

1. Explain how your reporting process ensures quality data:
 - Accuracy
 - Completeness
 - Consistency
 - Timeliness
 - Security / Privacy

(Write **in your own words** how your reports provide “the right information, to the right people, at the right time, in the right format.”)

C. DAMA Sub-Domains

1. Explain **in your own words** how principles/policies/procedures apply to your reporting pipeline. **Do not copy theory and do not restate DAMA definitions.**
2. **Map the ETL/reporting steps to DAMA sub-domains:** choose **three** relevant sub-domains from the DAMA Wheel aligned with the ETL/reporting workflow.
3. Define principles that guide each sub-domain and write **in your own words** the *corresponding* policies and procedures.

Note: each procedure should explain how it governs the creation of the operational and executive reports on a regular basis.

For each sub-domain:

Sub-domain 1: [Name]

- Principle 1
- Policy 1
- Procedure 1
- Principle 2
- Policy 2
- Procedure 2

Sub-domain 2: [Name]

- Principle 1
- Policy 1
- Procedure 1
- Principle 2
- Policy 2
- Procedure 2

Sub-domain 3: [Name]

- Principle 1
 - Policy 1
 - Procedure 1
 - Principle 2
 - Policy 2
 - Procedure 2
-

D. Governance Roles & Responsibilities

1. Identify **at least** three roles in the reporting process and their responsibilities:
 - Data Owner (e.g., accountable for data domain integrity)
 - Data Steward (e.g., ensures compliance, maintains metadata, validates data quality)
 - Report Consumer (e.g., executives, managers, operational staff using reports)

E. Risk Mitigation

1. Identify two key risks if governance is not applied (e.g., inconsistent KPIs, privacy breaches, misinterpretation of data) and describe how your framework reduces or prevents them.
 2. Quantify or prioritize the risks (high/medium/low)
-

F. Conclusion

1. ***In your own words*** summarize how the governance framework ensures that the reporting process is:
 - Transparent
 - Efficient
 - Quality-driven
 - Secure and compliant
-

Deliverables

- ✓ **Submit as a single PDF named: Group#_Project.pdf**
Include a cover page with group number and team contributors.
-

Annex I - Example of an Operational Report

**Sales Team Operations by Quarter
01/01/22 - 03/31/22**

Regions	Province	City	Sales per person	Item	Unit Price (\$)	Quantity Sold	Quantity Quota	Quantity Sold / Quantity Quota (%)	Discount Cap (%)	Gross Sales (\$)	Gross Sales / Gross Sales Quota (%)
Atlantic	New Brunswick	Moncton	S1	Chair	350	50.00	45.00	11%	10%	\$15,750	0%
				Desk	750	28.00	30.00	-7%	5%	\$22,050	-2%
				Bookshelf	1,200	32.00	30.00	7%	7%	\$36,480	1%
		Subtotal								\$74,280	0%
	Newfoundland and Labrador	St. John's	S2	Chair	350	30.00	30.00	0%	10%	\$9,450	-10%
				Desk	750	18.00	20.00	-10%	5%	\$13,905	-7%
				Bookshelf	1,200	10.00	16.00	-38%	7%	\$11,400	-41%
		Subtotal								\$34,755	-22%
	Nova Scotia	Halifax	S3	Chair	350	50.00	45.00	11%	10%	\$15,750	0%
				Desk	750	35.00	30.00	17%	10%	\$28,088	25%
				Bookshelf	1,200	32.00	30.00	7%	10%	\$36,480	1%
		Subtotal								\$80,318	8%
	Prince Edward Island	Charlottetown	S4	Chair	350	40.00	30.00	33%	10%	\$12,600	20%
				Desk	750	20.00	25.00	-20%	4%	\$15,000	-20%
				Bookshelf	1,200	10.00	10.00	0%	9%	\$11,400	-5%
		Subtotal								\$39,000	-5%
		Total								\$228,353	-3%
Central	Ontario	Toronto	S5	Chair	350	50.00	45.00	11%	10%	\$15,750	0%
				Desk	750	28.00	30.00	-7%	5%	\$22,050	-2%
				Bookshelf	1,200	32.00	30.00	7%	7%	\$36,480	1%
		Subtotal								\$74,280	0%
	Ottawa		S6	Chair	350	30.00	30.00	0%	10%	\$9,450	-10%
				Desk	750	18.00	20.00	-10%	5%	\$13,905	-7%
				Bookshelf	1,200	10.00	16.00	-38%	7%	\$11,400	-41%
		Subtotal								\$34,755	-22%
		Total								\$109,035	-8%
		Total								\$228,353	-3%
Grand Total											

Annex II - Example of an Executive Report

Sales Team Monthly Executive Report Gross Sales Revenue (\$)

Province	Jan 2022	Feb 2022	Feb 2022 vs Jan 2022	Feb 2021	Feb 22 vs Feb 21	Participation Ratio - 2021	Participation Ratio Feb 22
New Brunswick	\$51,250	\$56,000	9.3%	\$55,500	0.9%	30.0%	32.6%
Newfoundland and Labrador	\$32,560	\$35,000	7.5%	\$34,958	0.1%	25.0%	20.4%
Nova Scotia	\$65,900	\$68,456	3.9%	\$55,650	23.0%	35.0%	39.8%
Prince Edward Island	\$15,984	\$12,365	-22.6%	\$10,500	17.8%	10.0%	7.2%
East Region	\$165,694	\$171,821	3.7%	\$156,608	9.7%	100.0%	100.0%

Annex III – MySQL Workbench Installation

- **MySQL installation**

- Install MySQL V 8.X.X Win64 (x86_64) (MYSQL Community Server – GPL)
- Install MySQL Workbench 8.X - V 8.X Community

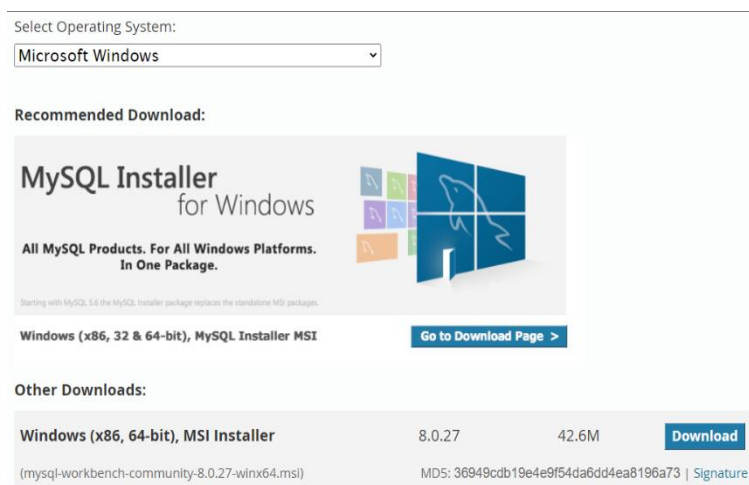
<https://dev.mysql.com/downloads/workbench/>

or

<https://dev.mysql.com/downloads/>

MySQL parameters

- Server: localhost
- Username: root
- Password: 1234
- Port 3306



Important

At the end of this project, **DO NOT** uninstall the software from your computers. Write down installation parameters, user IDs, passwords for later use. We will use these tools and deliverables again during the Data Visualization Techniques course.

Excerpt from the College Policy on Academic Dishonesty:

The minimal consequence for submitting a plagiarized, purchased, contracted, or in any manner inappropriately negotiated or falsified assignment, test, essay, project, or any evaluated material will be a grade of zero on that material. To view George Brown College policies please go to www.georgebrown.ca/policies